

KARAN PRATAP SINGH

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PROFESSIONAL SUMMARY

Highly accomplished Senior Software Engineer with extensive experience building high-throughput distributed systems, scalable data pipelines, and robust APIs. Strong proficiency in backend engineering with Python, Java, Go, and Redis. Proven track record of optimizing data ingestion streaming systems at a terabyte scale. Published author on System Design and Go, seeking to apply deep expertise in high-performance computing, data flow optimization, and distributed infrastructure to scale machine learning and data science workloads.

TECHNICAL SKILLS

Programming Golang, Python, TypeScript, JavaScript, Java, PostgreSQL, Redis, GraphQL, Frameworks Express.js, Django, Flask, FastAPI, Spring Boot, React.js, Next.js, DevOps Docker, Kubernetes, Terraform, Pulumi, Jenkins, Google Cloud Platform (GCP), google cloud, express, spring, golang, docker, react, gcp, aws

EXPERIENCE

Software Engineer 02/

05/2024 - Present

Apple Inc. Cupertino | CA

Senior Software Engineer 05/03/2023 | 01/26/2024

- Migrated legacy Java SOAP APIs to high-performance, robust RESTful APIs using Spring Boot, significantly improving backend service efficiency.
- Developed and optimized core features for web applications serving over 4M users using Node.js, ensuring high availability and seamless data flow.
- Enhanced system reliability and maintainability by increasing test coverage by 12% across critical application pathways.

Senior Engineer 12/07/2020 | 05/13/2023

Curebase San Francisco | CA

- Designed and implemented highly scalable REST and GraphQL APIs using Node.js, Redis, and PostgreSQL to handle high-concurrency client demands.
- Utilized Python, Golang (Go), and Terraform to develop business-critical internal tools, automating workflows and optimizing the software development lifecycle.
- Built fault-tolerant, horizontally scalable multi-tenant architectures leveraging AWS cloud services (ALB, ECS, S3, RDS, SQS, DynamoDB) with a focus on data security.
- Led codebase migrations and performance optimizations that reduced API latency and operational infrastructure costs by 50%.

Software Developer 05/26/2020 | 01/31/2021

Quotient Inc. Maryland | USA (Remote)

- Architected and implemented terabyte-scale data ingestion and streaming pipelines handling over 150 million artifacts with high throughput.
- Designed robust, end-to-end GraphQL data layers and efficient client-side enterprise search solutions.
- Optimized production CI/CD deployment pipelines using Docker multi-stage builds, reducing container footprint by 80% for streamlined cloud deployments.
- Created highly interactive digital asset management platforms using Node.js, TypeScript, and React.

Software Developer 08/17/2019 | 06/07/2020

Stewards London | UK (Remote)

- Engineered high-performance, event-driven backend systems and mobile applications utilizing modern JavaScript frameworks.
- Edge-optimized distributed GraphQL APIs utilizing Google Cloud CDN, enhancing response times and increasing customer retention by 60%.
- Consolidated internal data flows, sales tracking, and custom B2B integrations to improve operational workflows.

modeling". US Patent Application No. 18230253. Filed August 4 | 2023.

- "Learn Go", Published Jan 2023, ISBN 979-8-9883975-4-0.
- "System Design", Published Aug 2022, ISBN 979-8-9883975-8-8.
- ACHIEVEMENTS
- Co-authored U.S. patents, contributing novel solutions in software security and delivery pipelines.
- Author and maintainer of "System Design", ranked amongst GitHub's top 500 global repositories with
- over 38K stars and 40K monthly views.
- Published 50+ technical articles on software engineering topics such as distributed systems, microservices,
- software architecture, and DevOps on Medium and Dev.to, reaching over 2 million views.
- Created and managed a YouTube channel on software engineering, producing educational tutorials
- that have reached 70K+ developers globally.
- Published technical books on "System Design" and "Learn Go", contributing practical guidance and
- insights to the software engineering community.
- Contributed to open-source projects, acknowledged in React Native v0.60 by Meta.

EDUCATION

Bachelors of Technology, Computer Science and Engineering | SRM Institute of Science and Technology, Delhi NCR, India 2017 - 2021